

The smallest singular value of a sparse Mertens matrix

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Abstract

Redheffer's $(0, 1)$ matrix has determinant $M(n) = \sum_{k \leq n} \mu(k)$. We study a much sparser $(0, 1)$ matrix $\mathcal{R}_n$ with the same determinant: it has about $2.61n$ nonzero entries rather than order $n \log n$. We prove that its smallest singular value satisfies, unconditionally,

$$|M(n)| \cdot n^{-3/2+o(1)} \leq \sigma_{\min}(\mathcal{R}_n) \leq |M(n)| \cdot n^{-4/3+o(1)},$$

and that $\text{RH} \iff \sigma_{\min}(\mathcal{R}_n) \ll_{\varepsilon} n^{-1+\varepsilon}$. The equivalence is two-sided and unconditional. The key identity expresses the coordinates of $w = \mathcal{A}^{-\top}(\mathbf{1} - e_1)$, for the unit lower-triangular factor $\mathcal{A}$, as rough-Möbius sums $M(x, y) = \sum_{d \leq x, P^-(d) > y} \mu(d)$. Alladi's 1982 asymptotic for these sums [1] then gives $\|w\| = n^{1+o(1)}$. Under the precise rank-one dominance condition

$$\frac{|M(n)| \|\mathcal{A}_n^{-1}\|}{\|\mu_n\| \|w\|} \longrightarrow 0,$$

which follows from RH, the bracket sharpens to $\sigma_{\min}(\mathcal{R}_n) = |M(n)|n^{-3/2+o(1)}$. At the other end we prove $\sigma_{\max}(\mathcal{R}_n) = \sqrt{n}(1 + O(1/n))$ and $\sigma_{\max}/|\lambda_{\max}| \asymp \sqrt{\log n}$. We also prove $\|\mathcal{A}_n^{-1}\| = n^{1/2+o(1)}$, hence $\sigma_{\min}(\mathcal{A}_n) = n^{-1/2+o(1)}$, which answers a question of Cheon and Kim for this family. The new contribution is the linear-algebraic connection and its consequences; the sparse matrix, determinant identity, and analytic rough-Möbius estimate are prior work.

1 Introduction

This paper connects the smallest singular value of a sparse $(0, 1)$ matrix to the size of the Mertens function. Write μ for the Möbius function and

$$M(n) := \sum_{k \leq n} \mu(k)$$

for the Mertens function. The main result is the unconditional bracket

$$|M(n)|n^{-3/2+o(1)} \leq \sigma_{\min}(\mathcal{R}_n) \leq |M(n)|n^{-4/3+o(1)}$$

for the sparse matrix $\mathcal{R}_n$ defined below, together with the equivalence

$$\text{RH} \iff \sigma_{\min}(\mathcal{R}_n) \ll_{\varepsilon} n^{-1+\varepsilon}.$$

The proof identifies a vector in $\mathcal{R}_n^{-1}$ with Möbius sums over integers free of small prime factors and then applies a theorem of Alladi [1]. The matrix identity is proved here; the analytic estimate is prior work.

It is classical that the prime number theorem is equivalent to $M(n) = o(n)$, and that the Riemann hypothesis is equivalent to

$$M(n) = O(n^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0. \quad (1)$$

Redheffer [17] introduced the $n \times n$ $(0,1)$ matrix R_n with $R(i,j) = 1$ when $j = 1$ or $i \mid j$, and proved $\det R_n = M(n)$. This places (1) inside linear algebra, and a long line of work has studied the spectrum of R_n [4, 5, 18, 19, 7, 6].

Much less is known about *singular* values. Clément and Steinerberger [9] treat the largest singular value of Redheffer's matrix. Hilberdink [11, §4(b)] proves an $n^{-1/2}$ smallest-singular-value asymptotic for the dense divisibility triangular factor underlying Redheffer's matrix. That result is close in exponent but does not treat the sparse forest factor $\mathcal{A}_n$ or the modified matrix $\mathcal{R}_n$. The idea of routing the Riemann hypothesis through a smallest singular value appears to originate with Bordellès and Cloitre [3], who study an integer matrix Γ_n with $\det \Gamma_n = n! \sum_{k \leq n} \mu(k)/k$ and deduce from an LU decomposition that a lower bound $\sigma_{\min}(U_n) \gg_{\varepsilon} n^{-1/2-\varepsilon}$ would imply RH; Cheon and Kim [8] give a criterion of the same shape for the triangular factor of a Mertens equimodular matrix, and ask which matrices satisfy it. In both cases the criterion is a *sufficient condition* awaiting verification, and in neither is the relevant singular value determined. Here the modified matrix yields a two-sided equivalence, while the unmodified triangular factor fails the Cheon–Kim condition. These are statements about this particular sparse family, not a general singular-value criterion for Mertens matrices.

1.1 The matrix

The following much sparser $(0,1)$ matrix with the same determinant was introduced in [12] and restated in [13, §1]. Let $P(i)$ denote the largest prime factor of $i > 1$, and set

$$\mathcal{A}(i,j) := \begin{cases} 1 & \text{if } i = j, \\ 1 & \text{if } i \text{ is squarefree and } i/j = P(i), \\ 0 & \text{otherwise,} \end{cases} \quad \mathcal{R}_n := \mathcal{A}_n + e_1(\mathbf{1} - e_1)^{\top}, \quad (2)$$

so that $\mathcal{R}_n$ agrees with $\mathcal{A}_n$ except that its first row is all ones. Then $\mathcal{A}_n$ is unit lower triangular, $\det \mathcal{R}_n = M(n)$, and $\mathcal{R}_n$ has about $2.61n$ nonzero entries: one per diagonal, one per row of the all-ones row, and one for each squarefree $i \leq n$, the last contributing $6n/\pi^2 + O(\sqrt{n})$. By contrast

R_n has $\sum_{i \leq n} d(i) = n \log n + (2\gamma - 1)n + O(n^\theta)$ nonzeros. The two dominant eigenvalues of $\mathcal{R}_n$ are

$$r_{\pm} = 1 \pm \sqrt{\pi(n)} + \frac{1}{2} \frac{\pi'_2(n)}{\pi(n)} + O\left(\frac{\log \log^2 n}{\sqrt{n/\log n}}\right) \quad (3)$$

by [13, Thm. 1], where $\pi'_k(n)$ counts squarefree integers below n with exactly k prime factors.

The reason $\sqrt{\pi(n)}$ appears in (3), rather than the $\sqrt{n}$ of the Redheffer matrix [5, Thm. 1], is worth stating because the same mechanism governs $\sigma_{\max}$ below. Writing $\mathcal{A} = I + S$, the matrix S is the adjacency matrix of a *forest*: each squarefree i has the unique parent $i/P(i)$. The children of the root 1 are exactly the primes, so the hub degree is $\pi(n)$; in R_n the corresponding hub degree is n .

1.2 Results

Throughout, $P^-(i)$ is the smallest prime factor of $i > 1$ with $P^-(1) = \infty$, and

$$M(x, y) := \sum_{\substack{d \leq x \\ P^-(d) > y}} \mu(d) \quad (4)$$

is the Möbius function summed over *y-rough* integers. We write $\alpha = \log x / \log y$ and let ρ denote Dickman's function, defined by $\rho(\alpha) = 1$ on $0 < \alpha \leq 1$ and $\rho(\alpha) = 1 - \int_1^\alpha \rho(t-1) dt/t$ for $\alpha > 1$.

Set $\mu_n := (\mu(1), \dots, \mu(n))^T$ and

$$w := \mathcal{A}_n^{-T}(1 - e_1). \quad (5)$$

The first of these is the first column of $\mathcal{A}_n^{-1}$; the vector w collects the *other* columns, and it is the object that was missing from the analysis in [14].

Our first result identifies w completely.

Theorem 1 (closed form). *For $1 \leq j \leq n$,*

$$w_j = \begin{cases} 1 & \text{if } j \text{ is not squarefree,} \\ M(n/j, P(j)) & \text{if } j \text{ is squarefree,} \end{cases}$$

with the convention $P(1) := 1$, so that $M(n/1, P(1)) = M(n)$, and with the value at $j = 1$ diminished by 1. Consequently

$$\|w\|^2 = \#\{j \leq n : j \text{ not squarefree}\} + (M(n) - 1)^2 + \sum_{\substack{2 \leq j \leq n \\ j \text{ squarefree}}} M(n/j, P(j))^2.$$

So an inverse-matrix vector for $\mathcal{R}_n$ is, entry by entry, a rough-Möbius sum evaluated along the hyperbola $x = n/j$, $y = P(j)$. Alladi's asymptotic for $M(x, y)$ then controls $\|w\|$:

Theorem 2 (norm). $\|w\| = n^{1+o(1)}$, unconditionally.

Combining with an exact Sherman–Morrison factorisation gives the main result.

Theorem 3 (smallest singular value). *Unconditionally,*

$$|M(n)| \cdot n^{-3/2+o(1)} \leq \sigma_{\min}(\mathcal{R}_n) \leq |M(n)| \cdot n^{-4/3+o(1)},$$

the upper bound using the classical quantitative prime number theorem (Remark 13). If moreover

$$\frac{|M(n)| \|\mathcal{A}_n^{-1}\|}{\|\mu_n\| \|w\|} \rightarrow 0,$$

the two collapse to $\sigma_{\min}(\mathcal{R}_n) = |M(n)| \cdot n^{-3/2+o(1)}$. In either case

$$RH \iff \sigma_{\min}(\mathcal{R}_n) \ll_{\varepsilon} n^{-1+\varepsilon}.$$

The equivalence is two-sided and unconditional. The implication “ $\Leftarrow$ ” is the unconditional lower bound rearranged. For “ $\Rightarrow$ ”, RH makes the displayed rank-one dominance condition hold and hence supplies the exact asymptotic. The simpler condition $|M(n)|/\|w\| \rightarrow 0$ is not used as a substitute: the proof also has to control $\|\mathcal{A}_n^{-1}\|/\|\mu_n\| = n^{o(1)}$.

We stress what is and is not content here. That $\det \mathcal{R}_n = M(n)$ is [12, 13], so “RH $\iff |\det \mathcal{R}_n| \ll n^{1/2+\varepsilon}$ ” is immediate and carries nothing new. The content of Theorem 3 is the conversion factor $\|\mu_n\| \|w\| = n^{3/2+o(1)}$, that is, Theorem 2: without it the forward implication, and hence the full equivalence, is unavailable at the stated exponent.

At the other extreme, and for the unmodified factor:

Theorem 4 (the other extreme). $\sigma_{\max}(\mathcal{R}_n) = \sqrt{n}(1 + O(1/n))$, and $\sigma_{\max}(\mathcal{R}_n)/|\lambda_{\max}(\mathcal{R}_n)| \asymp \sqrt{\log n}$. Moreover $\|\mathcal{A}_n^{-1}\| = n^{1/2+o(1)}$ and hence $\sigma_{\min}(\mathcal{A}_n) = n^{-1/2+o(1)}$.

The exponent in the second assertion is sharp, but the $o(1)$ is an artefact of the method: the terminating Neumann series bounds $\|\mathcal{A}_n^{-1}\|$ by $\sum_r \sqrt{\pi'_r(n)}$, and that sum exceeds $\sqrt{n}$ by a factor 1.55, 1.62, 1.67, 1.72, 1.76 at $n = 10^4, \dots, 10^8$ — monotone increasing, so the argument cannot deliver $\asymp \sqrt{n}$ however carefully it is run. Numerically $\|\mathcal{A}_n^{-1}\|/\sqrt{n}$ is nearly constant over the dense-matrix range reported in §7, suggesting $\|\mathcal{A}_n^{-1}\| \asymp \sqrt{n}$; we do not prove that sharper estimate.

The last assertion answers a question of [8] for this family; see Remark 15.

1.3 Relation to earlier work

In [14] it is shown that for unit lower-triangular L , with B obtained by deleting the first row and l the first column of L^{-1} , one has $\det BB^{\top} = \|l\|_2^2$; for the matrices considered here $l = \mu_n$ and $\sqrt{\det BB^{\top}} = \sqrt{6n}/\pi + O(1)$. The same paper observes, informally, that

the Riemann hypothesis is an assertion that the constant $\mathbf{1}^{\top}$ is, for all n , almost in the span of the rows of B .

That distance is $|\det \mathcal{R}_n|/\sqrt{\det BB^\top} = |M(n)|/\|\mu_n\| \asymp |M(n)|n^{-1/2}$. Under the rank-one dominance condition of Theorem 3, the comparison becomes

$$\underbrace{\text{dist}(\mathbf{1}^\top, \text{rowspan } B)}_{= |M(n)|/\|\mu_n\| \asymp |M(n)|n^{-1/2}} \quad \text{versus} \quad \underbrace{\sigma_{\min}(\mathcal{R}_n)}_{= |M(n)|/(\|\mu_n\|\|w\|) = |M(n)|n^{-3/2+o(1)}}. \quad (6)$$

The first is the distance from one particular vector to a hyperplane; the second is the minimum over all unit directions. In the dominance regime they differ by $\|w\| = n^{1+o(1)}$. The unconditional bracket retains the same lower exponent but does not assert this exact comparison at every n .

1.4 What is cited and what is proved

The analytic engine is not ours. Theorem 5 below is [1, Thm. 1], together with the contrast with $\Psi(x, y)$ that he draws on p. 87 and that we record in Remark 6; [2] is its companion. What is established here is the linear algebra: Theorem 1, which identifies w with Alladi's sums, and the consequences 2–4. All numerical claims are reproduced by the scripts in `code/`. The bounded literature record and the scope of earlier AI-assisted checks are recorded in `audit/reports/`; those checks are not peer review.

2 Notation and cited results

For an arithmetic function f we write $A(f)_n$ for the $n \times n$ matrix with (i, j) entry $f(i/j)$ when $j \mid i$ and 0 otherwise. All matrices below are real. We use $\|\cdot\|$ for the spectral norm and $\sigma_1 \geq \dots \geq \sigma_n$ for singular values, so $\sigma_{\min} = \sigma_n$.

Recall $\mathcal{A} = I + S$ from (2), where

$$S(i, j) = 1 \iff i \text{ squarefree, } i \geq 2, j = i/P(i). \quad (7)$$

Each row of S has at most one nonzero entry, so S is the adjacency matrix of a forest in which every squarefree $i \geq 2$ has the single parent $i/P(i)$.

We use the following two results without proof.

Theorem 5 (Alladi [1, Thm. 1]). *If $y \geq x$ then $M(x, y) = 1$. If $y = x^{1/\alpha}$ then*

$$M(x, y) = \frac{x \rho'(\alpha)}{\log y} + \frac{y}{\log y} + O\left(\frac{x \alpha^2}{\log^2 y}\right),$$

uniformly for $2 \leq y < x$.

Differentiating the defining relation for ρ gives $\alpha \rho'(\alpha) = -\rho(\alpha - 1)$, so Theorem 5 may be written in the equivalent form

$$M(x, x^{1/\alpha}) = -\frac{x}{\log x} \rho(\alpha - 1) (1 + O_\alpha(1/\log y)), \quad (8)$$

which is the shape used below. Note $\rho > 0$ everywhere, so $M(x, y)$ is negative and bounded away from 0 relative to $x/\log x$ for each fixed α .

Remark 6. Alladi states the contrast with the smooth-number count $\Psi(x, y) \sim x\rho(\alpha)$: the same sum weighted by μ produces Dickman's *derivative* ρ' in place of ρ itself [1, p. 87]. It is worth adding, though it is not Alladi's remark, that the unweighted rough count $\Phi(x, y) = \#\{d \leq x : P^-(d) > y\} \sim x\omega(\alpha)/\log y$ involves Buchstab's ω ; so the μ -weighting replaces ω by ρ' .

Theorem 7 ([14, Thms. 1 and 3]). *Let L be unit lower triangular, let B be L with its first row deleted, and let l be the first column of L^{-1} . Then $\det BB^\top = \|l\|_2^2$. For $L = \mathcal{A}_n$ one has $l = \mu_n$ and $\|\mu_n\|^2 = \sum_{i \leq n} |\mu(i)| = 6n/\pi^2 + O(\sqrt{n})$.*

3 The inverse, and the vector w

Proof of Theorem 1. Since S is nilpotent, $\mathcal{A}^{-1} = \sum_{r \geq 0} (-S)^r$, and $(S^r)_{kj}$ counts paths of length r from k to j in the parent forest. Each vertex has at most one parent, so there is at most one such path, and it exists precisely when k is squarefree and j is obtained from k by repeatedly deleting largest prime factors. Writing k squarefree with primes $p_1 < \dots < p_r$, the vertices reachable from k are exactly the initial products $p_1 \dots p_m$, $0 \leq m \leq r$. Hence $(\mathcal{A}^{-1})_{kj} \neq 0$ if and only if k is squarefree, $j \mid k$, and every prime factor of k/j exceeds $P(j)$ — which forces j squarefree as well — and in that case $(\mathcal{A}^{-1})_{kj} = (-1)^{\omega(k/j)} = \mu(k/j)$.

Now $w_j = \sum_{k \geq 2} (\mathcal{A}^{-1})_{kj}$. If j is not squarefree the only surviving term is the diagonal one, giving $w_j = 1$. If j is squarefree, substitute $d = k/j$: the conditions become $d \leq n/j$, d squarefree, and $P^-(d) > P(j)$, whence $w_j = \sum_{d \leq n/j, P^-(d) > P(j)} \mu(d) = M(n/j, P(j))$; for $j = 1$ the constraint $k \geq 2$ removes the term $d = 1$. $\square$

Two consequences are worth isolating, because they make part of the sum in Theorem 1 completely explicit. Write $x = n/j$ and $y = P(j)$.

Corollary 8. *For squarefree j : if $y \geq x$ then $w_j = 1$; if $\sqrt{x} \leq y < x$ then $w_j = 1 - (\pi(x) - \pi(y))$.*

Proof. If $y \geq x$ the only $d \leq x$ with $P^-(d) > y$ is $d = 1$. If $\sqrt{x} \leq y < x$ then such a d is either 1 or a prime in $(y, x]$, since a composite y -rough number is at least $y^2 \geq x$. $\square$

The following two identities for $M(x, y)$ are standard and are used below; both are immediate from unique factorisation.

Lemma 9. *$M(x, y) = M(x) + \sum_{p \leq y} M(x/p, p)$, equivalently $M(x, y) = 1 - \sum_{y < p \leq x} M(x/p, p)$; and $M(x, y) = \sum_{a \leq x, a \text{ } y\text{-smooth}} M(x/a)$, the sum being over all y -smooth a , not merely squarefree ones.*

4 The norm of w

Proof of Theorem 2. Upper bound. Trivially $|M(x, y)| \leq x$, so by Theorem 1 $\|w\|^2 \leq \sum_{j \leq n} (n/j)^2 \leq \zeta(2)n^2$. Therefore $\|w\| \leq (\pi/\sqrt{6})n$.

Lower bound. Fix $\alpha > 1$ and let $j = p$ run over the primes in $(n^{1/(\alpha+1)}, 2n^{1/(\alpha+1)})$. For such p we have $x = n/p$, $y = p$ and $\log x / \log y = \alpha + O(1/\log n)$, so Theorem 5 applies at fixed α : its main term is $x\rho'(\alpha)/\log y \asymp_\alpha x/\log y$ and dominates the error $O(x\alpha^2/\log^2 y)$ by a factor $\log y$; the other main term $y/\log y$ is lower order because $y/x = n^{-(\alpha-1)/(\alpha+1)+o(1)}$. Since $\rho > 0$, $|w_p| = |M(n/p, p)| \gg_\alpha n^{\alpha/(\alpha+1)}/\log n$. By the prime number theorem there are $\gg n^{1/(\alpha+1)}/\log n$ such primes, and by Theorem 1 the quantity $\|w\|^2$ is a sum of squares, so *any* subfamily bounds it below:

$$\|w\|^2 \gg_\alpha \frac{n^{1/(\alpha+1)}}{\log n} \cdot \frac{n^{2\alpha/(\alpha+1)}}{\log^2 n} = \frac{n^{(2\alpha+1)/(\alpha+1)}}{\log^3 n}, \quad \text{i.e.} \quad \|w\| \gg_\alpha \frac{n^{1-\frac{1}{2\alpha+2}}}{\log^{3/2} n}.$$

Letting $\alpha \rightarrow \infty$ gives $\|w\| \gg_\varepsilon n^{1-\varepsilon}$ for every $\varepsilon > 0$. $\square$

Remark 10 (the refined shape, open). Both a saddle-point analysis and the numerics of §7 indicate

$$\|w\| = n \exp(-(1 + o(1))\sqrt{\log n \log \log n}),$$

with the maximising subfamily at $\alpha \asymp \sqrt{\log n / \log \log n}$. This is *not* established. Theorem 5 does not reach that range: as Alladi notes [1, p. 87], since $\rho'(\alpha) = -\exp\{-\alpha \log \alpha - \alpha \log \log \alpha + O(\alpha)\}$ for $\alpha > 3$, “the main terms of Theorem 1 are smaller than the error term when α is large”. His Theorem 2 covers large α but only as an upper bound with decay $\exp\{-(\alpha/2) \log \alpha\}$, half the true exponent. Determining the constant therefore requires an improvement of the 1982 estimates, and would sharpen the $o(1)$ in Theorems 2 and 3.

5 The smallest singular value

Proposition 11. $\mathcal{R}_n^{-1} = \mathcal{A}_n^{-1} - \mu_n w^\top / M(n)$ whenever $M(n) \neq 0$. If moreover

$$\frac{|M(n)| \|\mathcal{A}_n^{-1}\|}{\|\mu_n\| \|w\|} \longrightarrow 0, \tag{9}$$

then

$$\sigma_{\min}(\mathcal{R}_n) \cdot \|\mu_n\| \cdot \|w\| = |M(n)| (1 + o(1)). \tag{10}$$

Here the limits are understood along indices with $M(n) \neq 0$. At an index with $M(n) = 0$, the matrix $\mathcal{R}_n$ is singular and the corresponding zero-valued identity is immediate.

Proof. Sherman–Morrison applied to $\mathcal{R}_n = \mathcal{A}_n + e_1(1 - e_1)^\top$ gives the stated inverse, using $\mathcal{A}_n^{-1}e_1 = \mu_n$ and $(1 - e_1)^\top \mathcal{A}_n^{-1} = w^\top$, together with $1 + (1 - e_1)^\top \mathcal{A}_n^{-1}e_1 = M(n)$, which is $\det \mathcal{R}_n = M(n)$.

The rank-one term has norm $\|\mu_n\| \|w\| / |M(n)|$, while $\|\mathcal{A}_n^{-1}\| = n^{1/2+o(1)}$ by Theorem 4. Condition (9) says exactly that the norm of $\mathcal{A}_n^{-1}$ is negligible relative to the rank-one term. The reverse triangle inequality therefore gives $\|\mathcal{R}_n^{-1}\| = \|\mu_n\| \|w\| / |M(n)| (1 + o(1))$, and $\sigma_{\min}(\mathcal{R}_n) = 1 / \|\mathcal{R}_n^{-1}\|$ gives (10). $\square$

RH supplies (9): for any fixed $0 < \varepsilon < 1/2$ its left-hand side is $n^{-1/2+\varepsilon+o(1)}$. The condition is not known unconditionally. This is why Theorem 3 gives an unconditional bracket rather than an unconditional asymptotic.

Proof of Theorem 3. If $M(n) = 0$ then $\mathcal{R}_n$ is singular and the two bounds are immediate, so assume $M(n) \neq 0$. By Theorem 7, $\|\mu_n\| = \sqrt{6n/\pi^2} (1 + o(1)) = n^{1/2+o(1)}$, and by Theorem 2, $\|w\| = n^{1+o(1)}$.

Lower bound. From the inverse formula and the triangle inequality,

$$\sigma_{\min}(\mathcal{R}_n) \geq \frac{|M(n)|}{|M(n)| \|\mathcal{A}_n^{-1}\| + \|\mu_n\| \|w\|}.$$

The trivial bound $|M(n)| \leq n$, Theorem 4, and the elementary bounds $\|w\| \leq (\pi/\sqrt{6})n$ and $\|\mu_n\| \leq \sqrt{n}$ make the denominator $n^{3/2+o(1)}$. Thus $\sigma_{\min}(\mathcal{R}_n) \geq |M(n)| n^{-3/2+o(1)}$ unconditionally. *Upper bound.* Remark 13 gives $\sigma_{\min}(\mathcal{R}_n) \ll |M(n)| n^{-4/3+o(1)}$ from the classical zero-free-region bound for the Mertens function. If (9) holds, Proposition 11 applies and substituting the two norms into (10) collapses the bracket to $\sigma_{\min}(\mathcal{R}_n) = |M(n)| n^{-3/2+o(1)}$.

For the equivalence: if RH holds then $M(n) \ll n^{1/2+\varepsilon}$; together with $\|\mathcal{A}_n^{-1}\| = n^{1/2+o(1)}$, $\|\mu_n\| = n^{1/2+o(1)}$, and $\|w\| = n^{1+o(1)}$, this gives (9). The exact asymptotic is therefore available and $\sigma_{\min} \ll n^{-1+2\varepsilon}$. It is worth noting that the bracket alone does not give this: its upper bound combined with RH yields only $\sigma_{\min} \ll n^{1/2+\varepsilon} n^{-4/3+o(1)} = n^{-5/6+\varepsilon+o(1)}$, which is weaker than $n^{-1+\varepsilon}$. The forward implication genuinely needs the collapse, and hence Theorem 2. Conversely, $\sigma_{\min} \ll_{\varepsilon} n^{-1+\varepsilon}$ combined with the unconditional lower bound forces $M(n) \ll n^{1/2+\varepsilon}$, which is (1). Note that the reverse direction uses only the elementary upper bounds on $\|w\|$ and $\|\mu_n\|$; the forward direction is what Theorem 2 buys. $\square$

Remark 12. $\mathcal{R}_n$ is singular exactly when $M(n) = 0$. Thus no positive lower bound can hold at those indices; the lower side of Theorem 3 is scaled by $|M(n)|$. Test vectors and projections provide the complementary upper bounds; see Remark 13.

Remark 13 (a hierarchy of cheaper bounds). Assume $M(n) \neq 0$; when $M(n) = 0$ the asserted upper bounds are immediate. For any orthogonal projection Q one has $\|\mathcal{R}_n^{-1}\| \geq \|\mathcal{R}_n^{-1}Q\|$, and different choices of Q recover successively sharper bounds at successively higher cost. Taking the single test vector $\mu_n/\|\mu_n\|$ gives $\sigma_{\min} \leq |M(n)|/\|\mu_n\|$. Under (9), comparison with (10) shows that this is weaker by a factor $\|w\|(1 + o(1))$.

For an unconditional sharper bound, let Q project onto prime coordinates $p \in [n^{1/3}, 2n^{1/3}]$. Corollary 8 and the prime number theorem give

$$|w_p| = \pi(n/p) - \pi(p) - 1 = n^{2/3+o(1)}, \quad \|Qw\| = n^{5/6+o(1)}.$$

The closed form for $\mathcal{A}_n^{-1}$ shows that its column indexed by p is supported on squarefree k whose least prime factor is p . These supports are disjoint for distinct primes, and therefore

$$\|\mathcal{A}_n^{-1}Q\| = \max_p \sqrt{1 + \pi(n/p) - \pi(p)} = n^{1/3+o(1)}.$$

Using the inverse formula, the reverse triangle inequality, and the classical zero-free-region bound [16] $|M(n)| \ll n \exp(-c\sqrt{\log n})$, we obtain the explicit domination ratio

$$\frac{|M(n)| \|\mathcal{A}_n^{-1}Q\|}{\|\mu_n\| \|Qw\|} \ll \frac{|M(n)| \log n}{n} = o(1).$$

Consequently

$$\|\mathcal{R}_n^{-1}Q\| \geq \frac{\|\mu_n\| \|Qw\|}{|M(n)|} - \|\mathcal{A}_n^{-1}Q\| = \frac{n^{4/3+o(1)}}{|M(n)|}.$$

Thus $\sigma_{\min}(\mathcal{R}_n) \ll |M(n)| n^{-4/3+o(1)}$. The final step from $n^{-4/3}$ to the conditional $n^{-3/2}$ asymptotic requires the full estimate for $\|w\|$ from Theorem 5. Combining the unconditional upper bound with the same zero-free-region estimate gives $\sigma_{\min}(\mathcal{R}_n) \ll n^{-1/3+o(1)}$.

6 The largest singular value, and the unmodified factor

Proof of Theorem 4. Largest singular value. The first row of $\mathcal{R}_n$ has norm $\sqrt{n}$, so $\sigma_{\max} \geq \sqrt{n}$. For the upper bound write $\mathcal{R}_n = e_1 \mathbf{1}^\top + (I - e_1 e_1^\top) + S$. Each row of S has at most one nonzero entry and the column sums of S are the numbers of children, the largest being $\pi(n)$ at the root; the Schur test therefore gives $\|S\| \leq \sqrt{1 \cdot \pi(n)} = \sqrt{\pi(n)}$, and the star at the root shows this is an equality. The triangle inequality gives only $\sigma_{\max} \leq \sqrt{n} + 1 + \sqrt{\pi(n)} = \sqrt{n}(1 + O(1/\sqrt{\log n}))$, which is weaker than claimed; for the stated error term argue on the Gram matrix instead. Write

$$\mathcal{R}_n \mathcal{R}_n^\top = \begin{pmatrix} n & b^\top \\ b & C \end{pmatrix},$$

the first row of $\mathcal{R}_n$ being $\mathbf{1}^\top$. For $k \geq 2$, b_k is the number of nonzero entries in row k , which is 2 if k is squarefree and 1 otherwise; hence

$$\|b\|^2 = 4(Q(n) - 1) + (n - Q(n)) = 3Q(n) + n - 4 \leq 4n,$$

where $Q(n) = \#\{k \leq n : k \text{ squarefree}\}$. Also $C = BB^\top$ with B the last $n - 1$ rows of $\mathcal{R}_n$, so $\|C\| \leq (1 + \sqrt{\pi(n)})^2 = o(n)$. If $\lambda = \lambda_{\max}(\mathcal{R}_n \mathcal{R}_n^\top) \geq n$ then $\lambda > \|C\|$ for large n , and the Schur complement identity for the top-left entry gives

$$\lambda = n + b^\top (\lambda I - C)^{-1} b \leq n + \frac{\|b\|^2}{\lambda - \|C\|} = n + O(1).$$

Hence $\sigma_{\max} = \sqrt{n + O(1)} = \sqrt{n}(1 + O(1/n))$. (Numerically $\lambda_{\max}(\mathcal{R}_n \mathcal{R}_n^T) - n \rightarrow 1 + 18/\pi^2 = 2.8238$, which is $\lim \|b\|^2/n$, so the $O(1)$ is attained and not merely an upper bound.)

Non-normality. By (3), $|\lambda_{\max}(\mathcal{R}_n)| = (1 + o(1))\sqrt{\pi(n)}$, and $\sqrt{n/\pi(n)} \asymp \sqrt{\log n}$.

The unmodified factor. Let $\omega_{\max}(n)$ be the largest number of prime factors of a squarefree integer at most n , so $\omega_{\max}(n) \sim \log n / \log \log n$. Each application of S deletes one prime factor, hence $S^{\omega_{\max}(n)+1} = 0$ and the Neumann series $\mathcal{A}_n^{-1} = \sum_{r \leq \omega_{\max}(n)} (-S)^r$ terminates. Row sums of S^r are at most 1 and its column sums are $\pi'_r(n)$, the number of squarefree $k \leq n$ with $\omega(k) = r$; so $\|S^r\| \leq \sqrt{\pi'_r(n)}$ and $\|\mathcal{A}_n^{-1}\| \leq \sum_r \sqrt{\pi'_r(n)}$. There are at most $1 + \lfloor \log_2 n \rfloor$ nonzero summands and $\sum_r \pi'_r(n) \leq n$, so Cauchy–Schwarz gives $\sum_r \sqrt{\pi'_r(n)} \leq \sqrt{(1 + \lfloor \log_2 n \rfloor)n} = n^{1/2+o(1)}$. Conversely $\|\mathcal{A}_n^{-1}\| \geq \|\mathcal{A}_n^{-1} e_1\| = \|\mu_n\| \asymp \sqrt{n}$ by Theorem 7. Finally $\sigma_{\min}(\mathcal{A}_n) = 1/\|\mathcal{A}_n^{-1}\|$. $\square$

Remark 14 (Bordellès–Cloitre). The strategy of bounding a smallest singular value from below in order to reach PNT or RH is due to [3]. There U_n is the upper triangular factor of Γ_n , with $U_n^{-1} = (\theta_{ij})$ given explicitly in terms of iterated Möbius functions, and Corollary 2.7 of that paper states $|\sum_{k=i}^n \mu_i(k)/k| \leq 1/(n\sigma_n)$, so that $\sigma_n \gg_\varepsilon n^{-1+\varepsilon}$ suffices for PNT and $\sigma_n \gg_\varepsilon n^{-1/2-\varepsilon}$ suffices for RH. The authors observe that the general bounds available for smallest singular values of triangular matrices — for instance $\sigma_n \geq \min_i |a_{ii}|/2^{n-1}$ under a diagonal dominance hypothesis — are exponentially weak and “still very far from the PNT”.

Theorem 4 is a case where the arithmetic structure determines the exponent up to a subpolynomial factor: for the triangular factor $\mathcal{A}_n$ considered here, $\sigma_{\min}(\mathcal{A}_n) = n^{-1/2+o(1)}$, obtained not from a general matrix inequality but from the fact that the Neumann series for $\mathcal{A}_n^{-1}$ terminates after $\sim \log n / \log \log n$ terms. We note in passing that this is precisely the exponent appearing in the RH threshold of [3], though for a different matrix, so the two do not combine.

Remark 15 (a question of Cheon and Kim). For a Mertens equimodular matrix $L_n + uv^T$, since $\det L_n = 1$ and uv^T has singular values $\sqrt{n-1}, 0, \dots, 0$, the determinantal inequality of Li and Mathias [15] gives $|M(n)| \leq 1 + \sqrt{n-1}/\sigma_{\min}(L_n)$; Cheon and Kim [8, Thm. 4.2] deduce that $1 + \sqrt{n-1}/\sigma_{\min}(L_n) = O(n^{1/2+\varepsilon})$ implies the Riemann hypothesis, and ask which matrices satisfy this. Theorem 4 answers the question negatively for the present family: with $\sigma_{\min}(\mathcal{A}_n) = n^{-1/2+o(1)}$ the quantity is $n^{1+o(1)}$, which is not $O(n^{1/2+\varepsilon})$, so the resulting bound $|M(n)| \leq n^{1+o(1)}$ is trivial. Note their criterion concerns the *unmodified* factor and is one-directional, whereas Theorem 3 concerns the modified matrix and is an equivalence.

7 Numerical remarks

All values below are produced by the scripts in `code/`, using only NumPy (`checkB.py` additionally uses `mpmath`): the $O(n \log n)$ quantities by `wnorm.py` and `massprofile.py`, and every singular-value, eigenvalue and nilpotency figure by `spectra.py`, which builds the matrices explicitly and calls a dense SVD. Two independent implementations were used throughout: a dense construction with a direct linear solve, and an $O(n \log n)$ evaluator built from Theorem 1 and Corollary 8. They agree to residual exactly 0 at $n = 500, 1000, 2000$, which is the check on Theorem 1.

n	$\ w\ $	$\ w\ /n$	local exponent
10^5	2289.3	0.022893	—
$3 \cdot 10^5$	5211.8	0.017373	0.7488
10^6	12850.5	0.012851	0.7496
$3 \cdot 10^6$	29519.5	0.009840	0.7570
10^7	73972.3	0.007397	0.7630
$3 \cdot 10^7$	172026.3	0.005734	0.7683

Table 1: The norm $\|w\|$. The local exponent climbs steadily; Theorem 2 asserts it tends to 1.

Verification of (10), computed from the dense matrix: the ratio $\sigma_{\min}(\mathcal{R}_n)\|\mu_n\|\|w\|/|M(n)|$ equals 0.9896, 1.0038, 1.0047 at $n = 500, 1000, 2000$. Over the grid $n = 200, 225, \dots, 3200$ (excluding the n with $M(n) = 0$, where $\mathcal{R}_n$ is singular) it oscillates within $[0.987, 1.027]$, the extremes occurring at $n = 300$ and $n = 225$; the deviation is bounded and shows no trend, consistent with the $1 + o(1)$ of Proposition 11, but it is not monotone and does not visibly contract over this range. For $\|\mu_n\|$, the ratio $\|\mu_n\|/\sqrt{n}$ equals 0.77971 at $n = 10^5$ and 0.77970 at $n = 3 \cdot 10^5, 10^6, 3 \cdot 10^6, 10^7$, against $\sqrt{6/\pi^2} = 0.779697$: the squarefree count is already at its asymptotic density to five figures across this range.

For Theorem 4, at $n = 400, 800, 1600, 3200, 6400$ the values of $\sigma_{\max}(\mathcal{R}_n)/\sqrt{n}$ are

$$1.00387, \quad 1.00191, \quad 1.00094, \quad 1.00046, \quad 1.00023,$$

so the excess roughly halves at each doubling. At the same sizes, $\sigma_{\max}/|\lambda_{\max}|$ equals

$$1.906, \quad 2.087, \quad 2.262, \quad 2.439, \quad 2.592,$$

against $\sqrt{\log n} = 2.448, \dots, 2.960$. The nilpotency index of S is 5, 5, 5, 6 at $n = 500, \dots, 3000$; $\|\mathcal{A}_n^{-1}\|/\sqrt{n}$ equals 0.853, 0.851, 0.849, 0.850; and $\sigma_{\min}(\mathcal{A}_n)\sqrt{n}$ equals 1.1717, 1.1753, 1.1773, 1.1760. The quantity of Remark 15 equals 427, 851, 1699, 2552; for comparison, $\sqrt{n}$ equals 22, 32, 45, 55 at those sizes.

For scale, the same non-normality ratio computed for Redheffer's own R_n (built directly, and checked against $\det R_n = M(n)$) equals 1.2243, 1.2475, 1.2686, 1.2875 at $n = 400, 800, 1600, 3200$, against 1.9063, 2.0871, 2.2623, 2.4395 for $\mathcal{R}_n$. Both are far below $\sqrt{n}$, and at these sizes the sparser matrix is the more non-normal of the two; we draw no asymptotic conclusion from four points.

7.1 The problem is severely pre-asymptotic

We record this because computation up to $n = 10^7$ is actively misleading here, and several natural conjectures it suggests are false.

Table 1 makes $\|w\|$ look like a clean power of n : over $10^5 \leq n \leq 10^6$ the ratio $\|w\|/n^{3/4}$ is stable to four significant figures (0.40711, 0.40658, 0.40637). It is not a power. The ratio breaks

at $3 \cdot 10^6$ (0.40951) and the local exponent continues to climb past $3/4$, consistent with Theorem 2. Likewise, the normalisation $\|w\|(\log n)^{1.13}/n^{5/6}$ appears stable to under 1% per decade while in fact increasing monotonically at every step.

The mechanism is a *corner*. Writing $\beta = \log j / \log n$, the mass of $\|w\|^2$ concentrates at $\beta = 1/3$, equivalently $\alpha = 2$ — exactly the threshold of Corollary 8, below which the y -rough numbers under x are still only 1 and the primes. Dickman’s ρ has a corner at 1, so the relevant integrand has a jump in its derivative there, and the maximiser is *pinned* at $\beta = 1/3$ until $\log n$ exceeds an explicit threshold. The measured mass-mean β is 0.3368, 0.3364, 0.3361, 0.3351 at $n = 10^5, \dots, 10^7$: already drifting, and accelerating.

The practical consequence is that the exponent in Theorem 2 cannot be inferred numerically, and any conjecture about $\|w\|$ or $\sigma_{\min}(\mathcal{R}_n)$ based on data below 10^8 should be treated with suspicion.

8 Open problems

1. **The refined shape.** Establish $\|w\| = n \exp(-(1+o(1))\sqrt{\log n \log \log n})$, and thereby replace the $o(1)$ in Theorem 3 by an explicit function. By Remark 10 this requires extending Theorem 5 to $\alpha \asymp \sqrt{\log x / \log \log x}$ — the μ -weighted analogue of Hildebrand’s range for $\Psi(x, y) \sim x\rho(\alpha)$ [10] — or improving the decay in [1, Thm. 2] from $\exp\{-(\alpha/2) \log \alpha\}$ to $\exp\{-\alpha \log \alpha\}$.
2. **Other families.** By Remark 15 the present family fails the Cheon–Kim criterion because its triangular factor satisfies $\sigma_{\min}(\mathcal{A}_n) = n^{-1/2+o(1)}$. Is there a family in which $\sigma_{\min}(L_n) \gg n^{-\varepsilon}$? Here the first column of $\mathcal{A}_n^{-1}$ already supplies the lower bound $\|\mathcal{A}_n^{-1}\| \gg \sqrt{n}$; any candidate family must avoid an analogous obstruction. The same question for the matrix U_n of [3] appears to be open, and Theorem 4 suggests that a termination argument for the relevant Neumann series, rather than a general triangular-matrix inequality, is the tool to try.
3. **Deeper singular values.** (3) and Theorem 4 describe the extremes. The forest structure suggests that successive shells — squarefree integers with 2, then 3 prime factors — govern the next singular values, with $\pi'_2(n)$ already visible in (3).

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